Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/23/2010
PI Name:	Kramerich, George L. Ph.D.		
Project Title:	Study of the Application of Modern Theories of Interface Control and Ergonomics to Space Exploration Activities		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid transport phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	216-687-2405
Organization Name:	Cleveland State University		
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City:	Cleveland	**State:**	OH
Zip Code:	44115	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2006	**End Date:**	09/30/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Mullins, James M	**Contact Phone:**	216-433-8322
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC06AA26A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project is a Cooperative Agreement between NASA Glenn Research Center (GRC) and Cleveland State University (CSU) entitled “Study of the Application of Modern Theories of Interface, Control, and Ergonomics to Space Exploration Activities”. The ISS & Human Research Project Office (MAHO) at Glenn Research Center designs and builds experiments to be flown on the Space Shuttle and eventually on the International Space Station. Economy of space, weight, complexity, data storage, ergonomics, and many other factors present problems that also exist in industry. Many of the solutions can come from the same areas of study mentioned above. The main objectives of the effort are to research and investigate the use of the latest technologies, methods, techniques, etc. which pertain to control, interface, and ergonomics in industrial facilities and to examine their application to space exploration. In the MAHO High Bay Area, where construction of International Space Station facilities takes place, Cleveland State		

	is operating systems for the monitoring and ongoing study of activities and the environment, including the monitoring of air particle content, temperature, pressure, and humidity and the video monitoring and study of human activity in this environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Saving of money and time and improvement of activities in NASA scientific endeavors.
Task Progress:	The main objectives of the effort were to research and investigate the use of the latest technologies, methods, techniques, etc. which pertain to control, interface, and ergonomics in industrial facilities and to examine their application to space exploration. The work was done in large part at NASA Glenn Research Center, using selected facilities as real-world laboratories; such as Space Exploration projects including the Crew Exploration Vehicle (CEV), the Crew Launch Vehicle (CLV), and the ARESI-1; Space Station projects including the Fluid Combustion Facility (FCF); and test installations including the Payload Rack Checkout Unit (PRCU), and Thermal Test Facility (TTF). CSU was continuing its endeavors in the areas mentioned above when, near the end of 2009, we were officially notified that our funding would be terminated as of December 31, 2008. Work, therefore, became oriented toward the activities of winding down the project. Since Task was prematurely defunded on Jan., 2009 no progress was made since that time.
Bibliography Type:	Description: (Last Updated:)